

Increasing Logarithmic Weights for Rolling Validation: Theory

Aileen Tao

1 Introduction and research question

Ordinary rolling validation adds validation losses across successive validation times. Here later observations receive larger weights. The question is how this changes the expected advantage of one model, the variability of that comparison, and the probability of selecting the wrong model. In particular, we ask which forms of increasing logarithmic growth still allow weighted rolling validation to favor the oracle consistently. The companion report, *Increasing Logarithmic Weights for Rolling Validation: Simulation Study* (08B), examines the finite-sample behavior.

1.1 Statistical setting

Let $Y_i = \theta_\star + \xi_i$, where $\xi_i \stackrel{\text{iid}}{\sim} N(0, 1)$. At validation time i , compare Model 1, the oracle $\hat{\theta}_{i-1}^{(1)} = \theta_\star$, with Model 0, the fitted mean

$$\hat{\theta}_{i-1}^{(0)} = \frac{1}{i-1} \sum_{t=1}^{i-1} Y_t.$$

Model 1 uses the true parameter, while Model 0 estimates it from the observations available before time i . This comparison isolates the finite-sample cost of estimation in a simple setting.

1.2 Weighted rolling-validation statistic and selection criterion

For positive deterministic weights w_i , define

$$\Xi_n = \sum_{i=2}^n w_i \left[(Y_i - \hat{\theta}_{i-1}^{(0)})^2 - (Y_i - \hat{\theta}_{i-1}^{(1)})^2 \right]. \quad (1)$$

A positive Ξ_n means Model 0 has the larger weighted loss, so the oracle is selected. A negative Ξ_n favors Model 0 and is therefore an incorrect selection. Selection consistency means $\mathbb{P}(\Xi_n < 0) \rightarrow 0$.

We study each weight through three quantities: $\mathbb{E}[\Xi_n]$, the accumulated signal favoring the oracle; $\text{Var}(\Xi_n)$, the variability of that comparison; and their ratio $\text{Var}(\Xi_n)/\{\mathbb{E}[\Xi_n]\}^2$.

Expected signal. For any positive deterministic weights,

$$\mathbb{E}[\Xi_n] = \sum_{i=2}^n \frac{w_i}{i-1}. \quad (2)$$

This is the accumulated deterministic advantage of the oracle under the chosen weighting sequence.

Variability. Successive validation terms share training observations, so their dependence must be retained. The exact variance decomposition and its Gaussian orthogonality argument are given in Appendix A.

Chebyshev selection rule. Since $\mathbb{E}[\Xi_n] > 0$,

$$\mathbb{P}(\Xi_n < 0) \leq \frac{\text{Var}(\Xi_n)}{\{\mathbb{E}[\Xi_n]\}^2}. \quad (3)$$

Thus the common three-step calculation is to find the mean, find the variance, and test whether their Chebyshev ratio tends to zero. A vanishing ratio is sufficient for the wrong-selection probability to vanish.

1.3 Candidate weights

We compare four increasing weights with $w_i = 1$, which serves only as an unweighted baseline and is not part of the increasing-weight family.

Weight	Why it is included
1	Unweighted baseline
$\log(i + 1)$	Slowly increasing logarithmic weight
$[\log(i + 1)]^2$	Stronger logarithmic growth
$1 + \log(i + 1) + [\log(i + 1)]^2$	Polynomial-log case
$\log \log(i + e)$	Extremely slowly increasing case

Table 1: Weights considered in 08A and 08B, in their common order.

2 Positive log-power weights

The family $w_i = [\log(i + 1)]^p$, $p > 0$, provides one framework for different logarithmic growth rates. The exponent controls the relative emphasis on later validation observations. The question is whether the extra variability from increasing weights is dominated by the accumulated signal.

Theorem 2.1. *For every fixed $p > 0$,*

$$\text{Mean:} \quad \mathbb{E}[\Xi_n] \sim \frac{(\log n)^{p+1}}{p + 1},$$

$$\text{Variance:} \quad \text{Var}(\Xi_n) \sim 2(\log n)^{2p},$$

and

$$\text{Selection:} \quad \frac{\text{Var}(\Xi_n)}{\{\mathbb{E}[\Xi_n]\}^2} \sim \frac{2(p + 1)^2}{(\log n)^2} \longrightarrow 0.$$

Hence every fixed $p > 0$ gives selection consistency.

For every fixed $p > 0$, the ratio has order $(\log n)^{-2}$ and tends to zero. Weighting increases both signal and variance, but the squared mean grows two powers of $\log n$ faster. The full proof is given in Appendix B.

2.1 Two concrete choices

The log and squared-log weights are direct special cases.

Weight	$\mathbb{E}[\Xi_n]$	$\text{Var}(\Xi_n)$	Chebyshev ratio	Selection
$\log(i+1)$	$\sim (\log n)^2/2$	$\sim 2(\log n)^2$	$\sim 8/(\log n)^2$	Consistent
$[\log(i+1)]^2$	$\sim (\log n)^3/3$	$\sim 2(\log n)^4$	$\sim 18/(\log n)^2$	Consistent

Their finite-sample moments, ratios, and selection errors are examined in 08B, Sections 3–5.

3 Polynomial-log weights

The positive log-power result suggests that the dominant growth rate may determine the asymptotic behavior. This motivates a polynomial in $\log(i+1)$.

Theorem 3.1. *Suppose $w_i = P(\log(i+1)) > 0$, where $P(x) = cx^k + O(x^{k-1})$, $k \geq 1$, and $c > 0$. Then*

$$\text{Mean:} \quad \mathbb{E}[\Xi_n] \sim \frac{c}{k+1} (\log n)^{k+1},$$

$$\text{Variance:} \quad \text{Var}(\Xi_n) \sim 2c^2 (\log n)^{2k},$$

and

$$\text{Selection:} \quad \frac{\text{Var}(\Xi_n)}{\{\mathbb{E}[\Xi_n]\}^2} \sim \frac{2(k+1)^2}{(\log n)^2} \longrightarrow 0.$$

The lower-degree terms grow more slowly than the leading $c[\log(i+1)]^k$ term, so they do not change the leading mean or variance. In particular, $1 + \log(i+1) + [\log(i+1)]^2$ has the same leading asymptotics as $[\log(i+1)]^2$. See Appendix C for the proof and 08B, Sections 3–5 for the finite-sample comparison.

4 The log-log weight

The weight $w_i = \log \log(i+e)$ tests whether an even more slowly increasing sequence can still give a vanishing Chebyshev ratio.

Theorem 4.1. *For $w_i = \log \log(i+e)$,*

$$\text{Mean:} \quad \mathbb{E}[\Xi_n] \sim (\log n) \log \log n,$$

$$\text{Variance:} \quad \text{Var}(\Xi_n) \sim 2(\log \log n)^2,$$

and

$$\text{Selection:} \quad \frac{\text{Var}(\Xi_n)}{\{\mathbb{E}[\Xi_n]\}^2} \sim \frac{2}{(\log n)^2} \longrightarrow 0.$$

Even this extremely slow increase is sufficient in the present example. The proof is in Appendix D, and the matching finite-sample evidence is in 08B, Sections 3–5.

5 Theory summary

Weight	$\mathbb{E}[\Xi_n]$	$\text{Var}(\Xi_n)$	Chebyshev ratio	Selection
1	$\sim \log n$	$\rightarrow \pi^2$	$\sim \pi^2/(\log n)^2$	Consistent
$\log(i+1)$	$\sim (\log n)^2/2$	$\sim 2(\log n)^2$	$\sim 8/(\log n)^2$	Consistent
$[\log(i+1)]^2$	$\sim (\log n)^3/3$	$\sim 2(\log n)^4$	$\sim 18/(\log n)^2$	Consistent
$1 + \log(i+1) + [\log(i+1)]^2$	$\sim (\log n)^3/3$	$\sim 2(\log n)^4$	$\sim 18/(\log n)^2$	Consistent
$\log \log(i+e)$	$\sim (\log n) \log \log n$	$\sim 2(\log \log n)^2$	$\sim 2/(\log n)^2$	Consistent

Table 2: Theoretical predictions examined in the accompanying simulation report.

6 Conclusion

Every fixed positive log-power weight considered here remains selection consistent because its accumulated signal dominates its variability. For polynomial-log weights, the highest-degree term determines the leading behavior. The log-log result shows that even extremely slow increasing growth can be sufficient.

The signal must grow sufficiently fast relative to its standard deviation:

$$\frac{\sqrt{\text{Var}(\Xi_n)}}{\mathbb{E}[\Xi_n]} \rightarrow 0, \quad \text{equivalently} \quad \frac{\text{Var}(\Xi_n)}{\{\mathbb{E}[\Xi_n]\}^2} \rightarrow 0.$$

This signal-versus-noise condition is what makes the Chebyshev consistency argument work. The accompanying 08B report checks how clearly these asymptotic predictions appear at finite sample sizes.

Appendix

A General mean and variance decomposition

Define $S_m = \sum_{t=1}^m \xi_t$. Then

$$\Xi_n = \sum_{i=2}^n w_i \left\{ \frac{S_{i-1}^2}{(i-1)^2} - 2\xi_i \frac{S_{i-1}}{i-1} \right\}. \quad (4)$$

Because S_{i-1} is independent of ξ_i , $\mathbb{E}(\xi_i S_{i-1}) = 0$ and $\mathbb{E}(S_{i-1}^2) = i-1$, proving (2).

For $1 \leq t < n$ and $2 \leq l \leq n$, define

$$a_{n,t} = \sum_{i=t+1}^n \frac{w_i}{(i-1)^2}, \quad b_{n,l} = \sum_{i=l+1}^n \frac{w_i}{(i-1)^2} - \frac{w_l}{l-1}.$$

Expanding S_{i-1}^2 and collecting Gaussian monomials gives

$$\Xi_n - \mathbb{E}[\Xi_n] = \sum_{t=1}^{n-1} a_{n,t} (\xi_t^2 - 1) + 2 \sum_{l=2}^n b_{n,l} \sum_{k < l} \xi_k \xi_l. \quad (5)$$

The centered squares have variance 2, and each product $\xi_k \xi_l$ has variance 1. Distinct terms are uncorrelated because some independent centered Gaussian variable appears to an odd power in every mixed product. There are $l-1$ products with larger index l . Hence

$$\text{Var}(\Xi_n) = 2 \sum_{t=1}^{n-1} a_{n,t}^2 + 4 \sum_{l=2}^n (l-1) b_{n,l}^2. \quad (6)$$

For the increasing weights below, define

$$A_t = \sum_{i=t+1}^{\infty} \frac{w_i}{(i-1)^2}, \quad B_l = A_l - \frac{w_l}{l-1}, \quad T_n = A_n.$$

Then $a_{n,t} = A_t - T_n$ and $b_{n,l} = B_l - T_n$ exactly. The leading terms in B_l cancel.

B Proof for increasing log-power weights

Proof of Theorem 2.1. Replacing $[\log(i+1)]^p/(i-1)$ by $(\log i)^p/i$ changes the mean by a bounded amount. With $u = \log x$, integral comparison gives

$$\mathbb{E}[\Xi_n] \sim \int^n \frac{(\log x)^p}{x} dx = \frac{(\log n)^{p+1}}{p+1}.$$

For $I_p(t) = \int_t^\infty (\log x)^p x^{-2} dx$, integration by parts gives $I_p(t) = (\log t)^p/t + p I_{p-1}(t)$. Substituting $x = te^v$ shows $I_q(t) = O((\log t)^q/t)$ for fixed q . Sum-integral comparison therefore yields

$$A_t \sim \frac{(\log t)^p}{t}, \quad T_n \sim \frac{(\log n)^p}{n}. \quad (7)$$

To expose the cancellation in B_l , replace $(i-1)^{-2}$ in A_l by $[i(i-1)]^{-1}$; the error is $O((\log l)^p/l^2)$. Summation by parts gives

$$\sum_{i=l+1}^{\infty} \frac{w_i}{i(i-1)} = \frac{w_{l+1}}{l} + \sum_{i=l+2}^{\infty} \frac{w_i - w_{i-1}}{i-1}.$$

The first term differs from $w_l/(l-1)$ by $O((\log l)^p/l^2 + (\log l)^{p-1}/l^2)$, and the mean-value theorem gives $|w_i - w_{i-1}| \leq C_p(\log i)^{p-1}/i$. Thus

$$|B_l| \leq C_p \left\{ \frac{(\log l)^{p-1}}{l} + \frac{(\log l)^p}{l^2} \right\}. \quad (8)$$

In particular, $\sum_t A_t^2 < \infty$, so the diagonal part of (6) is bounded.

For the nonterminal off-diagonal part,

$$\sum_{l=2}^n (l-1)B_l^2 = O\left(\sum_{l=2}^n \frac{(\log l)^{2p-2}}{l}\right).$$

With $u = \log x$, the comparison integral is $\int^{\log n} u^{2p-2} du$. It is bounded for $0 < p < 1/2$, of order $\log \log n$ for $p = 1/2$, and of order $(\log n)^{2p-1}$ for $p > 1/2$. In every case,

$$\sum_{l=2}^n (l-1)B_l^2 = o((\log n)^{2p}). \quad (9)$$

The terminal contribution satisfies

$$T_n^2 \sum_{l=2}^n (l-1) \sim \frac{(\log n)^{2p}}{2}.$$

After expanding $(B_l - T_n)^2$, Cauchy–Schwarz bounds the cross term by

$$\begin{aligned} 2 \left| T_n \sum_{l=2}^n (l-1)B_l \right| &\leq 2 \left\{ \left(\sum_{l=2}^n (l-1)B_l^2 \right) \left(T_n^2 \sum_{l=2}^n (l-1) \right) \right\}^{1/2} \\ &= o((\log n)^{2p}). \end{aligned}$$

The bounded diagonal part, the B_l^2 part, and the cross term are lower order. Therefore

$$\text{Var}(\Xi_n) \sim 4T_n^2 \sum_{l=2}^n (l-1) \sim 2(\log n)^{2p}.$$

Dividing by the squared mean completes the proof. $\square$

C Polynomial-log extension

Proof of Theorem 3.1. Since $P(x) = cx^k + O(x^{k-1})$,

$$P(\log(i+1)) = c[\log(i+1)]^k + O([\log(i+1)]^{k-1}).$$

Integral comparison gives $\mathbb{E}[\Xi_n] = c(\log n)^{k+1}/(k+1) + O((\log n)^k)$. The tail argument above yields

$$A_t \sim \frac{c(\log t)^k}{t}, \quad B_l = O\left(\frac{(\log l)^{k-1}}{l} + \frac{(\log l)^k}{l^2}\right).$$

Hence $T_n \sim c(\log n)^k/n$, $\sum_{l=2}^n (l-1)B_l^2 = O((\log n)^{2k-1})$, and the cross term is lower order by Cauchy–Schwarz. The bounded diagonal term is also lower order, so

$$\text{Var}(\Xi_n) \sim 4T_n^2 \sum_{l=2}^n (l-1) \sim 2c^2(\log n)^{2k}.$$

□

D Log-log extension

Proof of Theorem 4.1. Integral comparison and $u = \log x$ give

$$\mathbb{E}[\Xi_n] = \int^n \frac{\log \log x}{x} dx + O(1) = (\log n) \log \log n - \log n + O(1).$$

The same tail comparison and summation by parts give

$$A_t \sim \frac{\log \log t}{t}, \quad B_l = O\left(\frac{1}{l \log l} + \frac{\log \log l}{l^2}\right).$$

Both $\sum_t A_t^2$ and $\sum_l (l-1)B_l^2$ converge, while $T_n \sim \log \log n/n$. The cross term is lower order by Cauchy–Schwarz. Thus

$$\text{Var}(\Xi_n) \sim 4T_n^2 \sum_{l=2}^n (l-1) \sim 2(\log \log n)^2,$$

which proves the result. □